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Stanford University
Postdoctoral Scholar
Civil and Environmental Engineering, Management Science & Engineering

Palo Alto, CA
2013 - 2015

EDUCATION

University of California, Berkeley
Ph.D. in Electrical Engineering and Computer Sciences

Berkeley, CA
Aug. 2008 - Dec. 2013

University of Toronto
Bachelor of Applied Science in Engineering Science

Toronto, ON, Canada
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PUBLICATIONS

Preprints

- [1] J. Anunrojwong and B. Zhang, *Residential battery pooling under backup commitments*, 2026.
- [2] A. Britto, T. Spina, W. Yang, S. Fowers, B. Zhang, and C. White, *Building power grid models from open data: A complete pipeline from openstreetmap to optimal power flow*, 2026. arXiv: 2605.04289 [eess.SY].
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- [9] Z. Ma and B. Zhang, *Controlling grid-connected inverters under time-varying voltage constraints*, 2025.
- [10] Y. Tang, J. Zhao, and B. Zhang, *Lyapunov function-based power system transient stability analysis considering inter-inverter coupling among grid following inverters*, 2025.

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- [2] L. Perkins and B. Zhang, “Resource allocation in electricity markets with budget-constrained customers,” *IEEE Control Systems Letters*, vol. 10, pp. 1009–1014, 2026.
- [3] Y. Tang, J. Zhao, and B. Zhang, “Transient stability assessment considering nonlinear couplings among grid following inverter-based renewable resources,” *Applied Energy*, 2026.
- [4] J. Li, M. Motoki, and B. Zhang, “Strategic and fair aggregator interactions in energy markets: Multi-agent dynamics and quasiconcave games,” *IEEE Transactions on Energy Markets, Policy and Regulation*, pp. 340–351, 2025.
- [5] L. Streitmatter, T. Joswig-Jones, and B. Zhang, “Geometry of the feasible output regions of grid-interfacing inverters with current limits,” *IEEE Control Systems Letters*, 2025.
- [6] Y. Jiang, H. Min, and B. Zhang, “Oscillations-aware frequency security assessment via efficient worst-case frequency nadir computation,” *Electric Power Systems Research*, vol. 234, p. 110 656, 2024.
- [7] J. Li, M. Motoki, and B. Zhang, “Socially optimal energy usage via adaptive pricing,” *Electric Power Systems Research*, vol. 235, p. 110 640, 2024.
- [8] J. Stiasny, B. Zhang, and S. Chatzivasilieiadis, “Pinnsim: A simulator for power system dynamics based on physics-informed neural networks,” *Electric Power Systems Research*, vol. 235, p. 110 796, 2024.
- [9] W. Cui, Y. Jiang, and B. Zhang, “Reinforcement learning for optimal frequency control: A lyapunov approach,” *IEEE Transaction on Power Systems*, vol. 38, no. 2, pp. 1676–1688, 2023.
- [10] W. Cui, W. Yang, and B. Zhang, “Predicting power system dynamics and transients: A frequency domain approach,” *IEEE Transaction on Power Systems*, 2023.
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Conference Papers

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- [2] T. Joswig-Jones, S. Dong, J. Tan, D. Wu, and B. Zhang, "A review of system strength metrics for inverter-based power systems," in *IEEE Power and Energy Society General Meeting (PESGM)*, 2026.

- [3] T. Joswig-Jones, S. Dong, B. Zhang, and J. Tan, “System strength sensitivity to power flow perturbations in ac power systems with inverter-based resources,” in *IEEE Power and Energy Society General Meeting (PESGM)*, 2026.
- [4] T. Joswig-Jones and B. Zhang, “Safe trajectory gradient flow control of a grid-interfacing inverter,” in *IEEE Power and Energy Society General Meeting (PESGM)*, 2026.
- [5] Y. Tang, J. Zhao, and B. Zhang, “Transient stability analysis of mixed gfm-gfl inverter system based on contraction theory,” in *IEEE Power and Energy Society General Meeting (PESGM)*, 2026.
- [6] N. Christianson, W. Cui, S. Low, W. Yang, and B. Zhang, “Fast and reliable $n - k$ contingency screening with input-convex neural networks,” in *Learning for Dynamics & Control Conference*, 2025.
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- [8] P. Zhang and B. Zhang, “Optimal voltage control using online exponential barrier method,” in *North American Power Symposium*, 2025.
- [9] H. Alduaij, B. Zhang, and Y. Weng, “Determining feasibility of dc power flow using graph neural networks,” in *2024 IEEE Power & Energy Society General Meeting (PESGM)*, 2024, pp. 1–5.
- [10] Y. Jiang, H. Min, and B. Zhang, “Oscillations-aware frequency security assessment via efficient worst-case frequency nadir computation,” in *Power Systems Computation Conference (PSCC)*, 2024.
- [11] T. Joswig-Jones and B. Zhang, “Optimal control of grid-interfacing inverters with current magnitude limits,” in *IEEE Conference on Decision and Control*, 2024.
- [12] T. Joswig-Jones and B. Zhang, “Safe control of grid-interfacing inverters with current magnitude limits,” in *HICSS*, 2024.
- [13] J. Li, M. Motoki, and B. Zhang, “Balancing fairness and efficiency in energy resource allocations,” in *IEEE Conference on Decision and Control*, 2024.
- [14] J. Li, M. Motoki, and B. Zhang, “Socially optimal energy usage via adaptive pricing,” in *Power Systems Computation Conference (PSCC)*, 2024.
- [15] Z. Ma and B. Zhang, “Controlling grid-connected inverters under time-varying voltage constraints,” in *IEEE Power and Energy Society General Meeting*, 2024.
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- [26] L. Zhang and B. Zhang, “An iterative approach to improving solution quality for ac optimal power flow problems,” in *ACM E-Energy*, 2022.
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- [72] Y. Kanoria, A. Montanari, D. Tse, and B. Zhang, “Distributed storage for intermittent energy sources: Control design and performance limits,” in *Communication, Control, and Computing (Allerton), 2011 49th Annual Allerton Conference on*, 2011, pp. 1310–1317.

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- [74] B. Zhang and M. D. Trott, “Reference-free audio matching for rendezvous,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2010, pp. 3570–3573.
- [75] B. Zhang, S. B. Kuntze, L. Pavel, and J. S. Aitchison, “Delay-tolerant control design for semiconductor optical amplifiers,” in *American Control Conference (ACC)*, 2008, pp. 4154–4160.
- [76] B. Zhang, E. Young, and C. Simmons, “3-d computational analysis of hemodynamic shear stresses on the aortic valve,” in *Canadian Congress of Applied Mechanics*, 2007.

Book Chapters

Y. Zhao and B. Zhang, “Deep Learning in Power Systems”, in *Advanced Data Analytics for Power Systems*, A. Tajer, S. M. Perlaza and H. V. Poor, Ed. Cambridge University Press, 2021

DISSERTATION

B. Zhang, “Control and optimization of power systems with renewables: Voltage regulation and generator dispatch,” Ph.D. dissertation, University of California, Berkeley, 2013

STUDENTS and POSTDOCS

Current Group Members

- Lauren Streitmatter
- Margaret Capetz
- Lila Perkins
- Jiayi Yao
- Ruonan (Pina) Pi
- Andrew Schallwig

Alumni in Academia

- Jerry Anunrojwong, Assistant Professor, Yale School of Management
- Wenqi Cui, Assistant Professor, New York University
- Zixiao Ma, Assistant Professor, Binghamton University
- Yan Jiang, Assistant Professor, Chinese University of Hong Kong, Shenzhen
- Yize Chen, Assistant Professor, University of Alberta
- Yuanyuan Shi, Assistant Professor, University of California at San Diego
- Bolun Xu, Assistant Professor, Columbia University
- Shreyas Sekar, Assistant Professor, Rotman School of Management, University of Toronto
- Hao Wang, Assistant Professor, Monash University
- Pan Li, Professor, Tongji University

Alumi in Industry

- Trager Joswig-Jones, Staff Scientist, Lawrence Livermore National Laboratory
- Jiayi Li, Uber
- Daniel Tabas, Scientist, Base Power Company
- Ling Zhang, Research Scientist, Microsoft Research Asia
- Atinuke Ademola-Idowu, Research Scientist, GE Research
- Chase Dowling, Engineering Lead, Base Power Company
- Jesus Elmer Contreras Ocaña, Quantitative Analyst, ENGIE

Mentorship and Outreach Activities

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- Working with community colleges in the Seattle area on curriculum design for a more seamless transition for transfer students to UW
 - Working with Northwest Indian College (NWIC, a tribal land-grant university) to construct a micro-grid, and create their engineering curriculum and research programs.
 - Member of UW working group on AI for environment and sustainability
 - Member for faculty advisory board for the Clean Energy Institute
 - ECE departmental awards coordinator
 - IEEE HKN faculty Advisor
 - Mentor in the College of Engineering STARS program (works with student in low-income, first-generation, and underserved backgrounds)
 - Mentor in Clean Energy Institute REU programs (Native American and first generation students)
 - Led demos at Pacific Science Center for K-12 students
 - Faculty mentor for multiple capstone projects
 - Working with Yakama Nation to improve their energy infrastructure

BEST PAPERS

Best Conference Papers on Emerging Technologies, Power and Energy Society General Meeting, “Safe Trajectory Gradient Flow Control of a Grid-Interfacing Inverter”, with Trager Joswig-Jones, 2026

Best Conference Papers on Power System Operation, Planning, and Economics, Power and Energy Society General Meeting, “A Review of System Strength Metrics for Inverter-based Power Systems”, with Trager Joswig-Jones, Shuan Dong, Jin Tan, Di Wu 2026

Best Conference Papers on Analytic Methods for Power System , Power and Energy Society General Meeting, “Voltage Regulation in Distribution Systems with Data Center Loads”, with Yize Chen, 2026

Best Paper Finalist, ACM E-Energy, “An iterative approach to improving solution quality for AC optimal power flow problems”, with Ling Zhang, 2022

Best Conference Papers on Renewables, Storage, and Electric Vehicles, Power and Energy Society General Meeting, “State-of-Charge Aware EV Charging”, with Yize Chen, 2022

Highlight Paper, Power Systems Computation Conference (PSCC), “Data-Driven Optimal Voltage Regulation Using Input Convex Neural Networks”, with Yize Chen and Yuanyuan Shi, 2020

Best Paper Runner Up, ACM E-Energy, “Exploiting Vulnerabilities of Load Forecasting Through Adversarial Attacks”, with Yize Chen and Yushi Tan, 2019

Best Paper Finalist, ACM E-Energy, “A capacity-price game for uncertain renewables resources”, with Pan Li and Shreyas Sekar, 2018

Prize Paper (4 out of 2000 submitted papers) in Power and Energy Society General Meeting for “Non-Wire Alternatives to Capacity Expansion”, with Jesus E. Contreras-Ocana and Uzma Siddiqi, 2018

Best Conference Papers on Power System Stability, Control, and Protection, Power and Energy Society General Meeting, “Optimal design of virtual inertia and damping coefficients for virtual synchronous machines”, with Atinuke Ademola-Idowu 2018

RESEARCH GRANTS

1. J. Choi, **B. Zhang**, Y. Weng, *Collaborative Research: Unified Modeling of Grid-Data Center Power Electronics Interactions for Reliable and Stable Power Systems*, National Science Foundation, 2026-2029, Amount \$500,000
2. M. Wu and **B. Zhang**, *From Reactive to Proactive Nodal Net Demand Uncertainty Mitigation: Prosumer-behavior-aware Prediction, Incentivization, Control, and Evaluation*, Encouraged by the Department of Energy, 2026-2029, Amount \$1,200,000
3. V. Sathe, S. Kumar, **B. Zhang**, *Collaborative Research: ASCENT: Optimal Thermal Management for Continued Scaling of 3D Heterogeneous Systems*, National Science Foundation, 2025-2029, Amount \$1,500,000
4. **B. Zhang**, J. Choi, J. Lukuyu, *Advancing Power and Energy Research*, Galloway Foundation, 2025-2028, Amount \$600,000
5. **B. Zhang**, J. Choi, J. Lukuyu, D. Kirschen, J. Yu, *Center Incentive Funding: Reimagining Loads in Power Systems*, University of Washington, 2025-2027, Amount \$250,000
6. J. Lukuyu and **B. Zhang**, *Climate and Equity Considerations for Community-Based Demand Response Strategies in the Pacific Northwest*, Washington Clean Energy Institute, 2023-2024, Amount \$100,000
7. **B. Zhang**, *CAREER: Optimal Control of Energy Systems via Structured Neural Networks: A Convex Approach*, National Science Foundation, 2020-2028, Amount: \$500,000
8. **B. Zhang**, Y. Shi (UCSD), G. Qu (CMU), *Collaborative Research: Data-driven Power Systems Control with Stability Guarantee: A Lyapunov Approach*, National Science Foundation, 2022-2025, Amount \$500,000 (share \$100,000)
9. **B. Zhang**, W. Yang (Microsoft Research), Y. Xu (Breakthrough Energy), *Learning Power System Dynamics in the Frequency Domain*, Climate Change AI Innovation Grants Program, 2022-2023, Amount \$ 145,000
10. V. Sathe, **B. Zhang**, *Exploiting Formal Optimization, and Robust Control for Secure Power Delivery Systems*, Intel, Amount \$50,000
11. **B. Zhang**, Y. Zhao (Stony Brook), *Collaborative Research: Learning for faster computations: Enhancing the Efficiency and Security of Power System Operations*, National Science Foundation, 2020-2023, Amount \$450,000 (share \$225,000)

12. **B. Zhang**, B. Johnson, D. Kirschen, *Enhanced Power System Stability using Fast, Distributed Power Electronics Control*, National Science Foundation, 2019-2022, Amount: \$400,000
13. B. Johnson, D. Kirschen, **B. Zhang**, *A Scalable Control Architecture for 100% PV Penetration with Grid Forming Inverters*, Department of Energy, UW Amount: \$3,000,000 (share \$400,000)
14. L. Ratliff, **B. Zhang**, R. Johari (Stanford), *CPS: Small: Collaborative Research: Information Design and Price Mechanisms in Platforms for Cyber-Physical Systems with Learning Agents*, National Science Foundation, 2019-2022, Amount: \$450,000 (share \$298,788)
15. **B. Zhang**, *Integrating Datacenter into the Grid Using Storage*, Microsoft, Gift, Amount: \$45,000
16. **B. Zhang**, Y. Weng (ASU), *Collaborative Research: Learning and Optimizing Power Systems: A Geometric Approach*, National Science Foundation, 2018-2021, Amount: \$450,000 (share \$225,000)
17. D. Kirschen, **B. Zhang**, L. Pozzo, C. Lo Prete (Penn State), *CRISP 2.0 Type 1: Collaborative Research: Economic Mechanisms for Grid Resilience Against Extreme Events and Natural Gas Disruptions*, National Science Foundation, 2018-2021, Amount: \$750,000 (share \$500,000)
18. **B. Zhang**, *Managing Complex Buildings Using Deep Neural Networks: Control and Predictive Maintenance*, Centrica, 2018, Amount: \$85,000
19. **B. Zhang**, *Integration of Battery Modeling with Solar Building Energy Storage*, Battery Infomatics, 2018, Amount: \$30,000
20. V. Sathe, **B. Zhang**, *SaTC: STARSS: Small: Design of Low-Power True Random Number Generator based on Adaptive Post-Processing*, National Science Foundation, 2017-2020, Amount: \$246,667
21. **B. Zhang**, D. Kirschen, *Understanding Storage Degradation in Electricity Markets*, Doosan GridTech, 2017, Amount: \$26,000
22. **B. Zhang**, L. Ratliff, A. Dubey (Vanderbilt), *US Ignite: Collaborative Research: Focus Area 1: Social Computing Platform for Multi-modal Transit*, National Science Foundation, 2016-2019, Amount: \$600,000 (share \$300,000)
23. **B. Zhang**, L. Ratliff, *EAGER: Congestion Mitigation via Better Parking: New Fundamental Models and A Living Lab*, National Science Foundation, 2016-2018, Amount: \$218,130
24. M. Ortega-Vazquez, **B. Zhang**, *Enabling Market Participation of Storages via Aggregation*, UW Clean Energy Institute, 2017-2018, Amount: \$25,904
25. D. Kirschen, M. Ostendorf, **B. Zhang**, *Supplement: EPCN: Quantifying the Resilience of Power System to Natural Disasters*, National Science Foundation, 2016-2017, Amount: \$69,994
26. **B. Zhang**, R. Johari (Stanford), *CPS: Breakthrough: Collaborative Research: The Interweaving of Humans and Physical Systems: A Perspective from Power Systems*, National Science Foundation, 2015-2018, Amount: \$500,000 (share \$250,000)
27. **B. Zhang**, V. Sathe, *Improving Power System Stability via Power Electronic Interfaces of Renewable Resources*, UW Clean Energy Institute, Duration: 2015-2016, Amount: \$54,152
28. R. Johari, R. Rajagopal and **B. Zhang**, *Efficiency and Group Behaviour in Power Distribution Networks*, The Precourt Institute for Energy Efficiency and the TomKat Center for Sustainable Energy, Stanford University, Duration: 2013-2014, Amount: \$150,000

TEACHING

- EE 351, Introduction to Energy Systems
- EE 451, Wind Energy
- EE 454, Power System Analysis
- EE 456, Computer-Aided Design in Power Systems
- EE 497/498, Entrepreneurial Capstone
- EE 552, Power System Dynamics and Control
- EE 553, Power System Economics
- EE 554, Large Scale Power Systems
- EE 559, Graphs and Networks
- PMP 559, Data Science for Power Systems

HONORS and AWARDS

Keith & Nancy Rattie Endowed Professor in Electrical Engineering, University of Washington
Outstanding Editor, IEEE Transactions on Energy Markets, Policy, and Regulation, 2024
ACM SIGEnergy Rising Star Award, 2022
NSF CAREER Award, 2020
Best Reviewer, IEEE Transactions on Smartgrid, 2018
Forbes Magazine 30-Under-30 in Energy, 2015
NSERC Post Graduate Fellowship, Government of Canada, 2008-2009, 2011-2013
EECS Departmental Fellowship, U.C. Berkeley, 2008-2009
Ontario Professional Engineers Foundation for Education Gold Medal, 2008

INDUSTRY EXPERIENCE

Consulting Faculty, Microsoft Research, 2026–present

Science Advisor, 4 Quadrant Networks, 2024–present

AI Advisor, Electravehicles, 2019–2023

Research Intern, ISO New England, Sep. – Dec. 2012:

Worked with Dr. Eugene Litvinov in the department of Business Architecture & Technology. We investigated how to model and solve the unit commitment problem under high penetration of renewable energies. We built a stochastic unit commitment model in the GAMS environment and performed statistical analysis of load, wind and interchange data to reduce the overall dimensionality of the problem. We also investigated new market and pricing mechanisms to understand the impact of uncertainty on the overall system and individual participants.

Research Intern, HP Labs, Jun. – Sep. 2009:

Worked with Dr. Mitch Trott. We considered the problem of efficiently determining whether multiple microphones are listening to the same acoustic signal. We cast the problem as a hypothesis testing problem, and argued that an effective test statistic is found by quantized frequency-domain inner products. This approach was validated with a set of real world speech samples.

PROFESSIONAL EXPERIENCES

Member of the Advisory Board, Electrical Engineering, UC Merced

Process Chair, Climate Change AI Fund, 2024

Editor, IEEE Transactions on Energy Markets, Policy and Regulation

General Chair, International Conference on Energy Conversion & Storage (ORCAS), 2022

Reviewer for IEEE Transaction on Power System, IEEE Transaction on SmartGrid, Operations Research, IEEE Transaction on Control of Networked Systems, IEEE Transaction on Automatic Control, Operations Research, SmartGridComm, CDC, ACC, ECCC

TPC members: ACM E-Energy, IEEE SmartgridComm, GlobalSip